Revisiting Sequence-To-Sequence Piano Transcription With Transformers

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Introduction

Our goal is to critically replicate "Sequence-to-Sequence Piano Transcription with Transformers" (Hawthorne et al., 2021) reimplementing the encoder-decoder transformer architecture that converts audio frames into MIDI-like sequences. This will enable rigorous evaluation of some of the claims of the original research and supplementing their ablation studies with our own. Furthermore, we will be able to revisit the overarching idea of the paper—that state-of-the-art results can be achieved by a generic Transformer model without domain-specific adaptations—at an even smaller scale of data & compute. We intend to accessibly demo the audio transcription enabled by the transformer, comparing it with the original training label in the form of real audio.

Literature Review

Automatic Music Transcription (AMT) has been dominated by increasingly complex, domain-specific neural architectures tailored to piano transcription. The Onsets and Frames model (Hawthorne et al., 2018) exemplified this trend with separate convolutional stacks for detecting note onsets, note presence, and velocities, requiring non-differentiable decoding to combine outputs. While Awiszus (2019) attempted a sequence-to-sequence approach with Transformers, it achieved limited success, which the authors attribute to using relative rather than absolute time shifts and training on the smaller MAPS dataset instead of MAESTRO. This trajectory of task-specific architectural complexity set the stage for a fundamental question: could state-of-the-art performance be achieved with a generic architecture instead?

Hawthorne et al. (2021) answer affirmatively, demonstrating that a generic encoder-decoder Transformer architecture adapted from T5 (Raffel et al., 2020) can achieve state-of-the-art piano

transcription without domain-specific modifications. The model encodes log mel spectrogram frames and autoregressively decodes a MIDI-like event vocabulary consisting of note-on/off tokens, velocity changes, and absolute time shifts. Critically, they employ fixed absolute positional embeddings rather than T5's logarithmically scaled relative positional bucket embeddings to ensure uniform attention resolution across all positions, and they use absolute time shifts (from segment start) instead of relative time shifts (from previous event) to avoid cumulative timing errors during decoding.

While Hawthorne et al.'s results represent an important advance, independent replication remains essential to validate their findings and deepen understanding of which architectural elements drive performance. Critical examination through reimplementation can both verify the reproducibility of their claims and probe the robustness of their design choices under different conditions.

Methodology

Our work critically replicates Hawthorne et al.'s (2021) sequence-to-sequence piano transcription system, implementing their encoder-decoder Transformer architecture in PyTorch. While the original study used JAX to train on the full MAESTRO V3.0.0 dataset (~962 training performances) for 400K steps with an effective batch size of 256, our implementation operates at a substantially reduced scale due to computational constraints. Accordingly, we trained on a subset of 20 training, 3 validation, and 3 test music pieces for approximately 11,000 steps with an effective batch size of 32. Otherwise, our work closely mimics the 8 encoder and 8 decoder layer, 54 million parameter architecture detailed in Table 3 of Hawthorne et al. (2021) in Pytorch, including in the preprocessing pipeline that transforms .wav and .mid files into continuous

spectrogram inputs and one-hot-encoded MIDI-like events as training examples. For details, see the original repository cited in the paper (github.com/magenta/mt3).

To stress test the original architecture beyond the aforementioned data/resource constraints, we supplemented the MAESTRO dataset—which is composed of real performances and used to train the model—with a second MAPS dataset of mostly synthesized audio. We handled the differing shapes and metadata of these two datasets in a pre-preprocessing step that normalizes the datasets and caches them for training. MAPS was used for tests & experiments, while we continued to train on MAESTRO as was done for the original model.

In our critical analysis of Hawthorne et al's architecture and claims, we experimentally verified two claims by comparison of transcription quality: that discluding log mel scaling after the spectrogram computation allows successful training but causes somewhat poorer performance; and secondly, that representing time as an absolute offset from the start of a segment rather than as a step-wise relative shift from the previous event produces substantially better transcription performance.

Beyond replicating these claims, we also wanted to probe two architectural decisions through ablation. The paper swaps T5's logarithmically scaled relative positional bucket embeddings for fixed absolute positional embeddings, arguing this gives uniform attention resolution across all positions; we hypothesized this choice is valuable and that reverting to relative bucket embeddings would meaningfully hurt performance. Conversely, we suspected that reducing the number of attention heads from 6 to 4 would have little to no effect at our unavoidably smaller training scale, since the degree of head specialization the original model achieves was likely made possible by their larger scale in data and training steps than ours. Both of these hypotheses

were also evaluated by comparison of transcription quality between the two configurations.

We evaluated transcription quality for these experiments using standard note-level precision and recall metrics, following the methodology used by Hawthorne et al. For each of our experiments, we report three progressively stricter evaluation criteria for generating our F1 scores: Onset-only matching, which considers only note start times; onset+offset matching, which requires both note boundaries to be correct; and onset+offset+velocity matching, which additionally verifies note loudness. In most cases, correctness was measured under a tolerance of 50 ms. The choice to use these particular metrics keeps our experiments comparable to previous works; this is the standard in AMT research.

Results

The replication succeeded in producing MIDI-like events that, when detokenized and played, were perceptually convincing duplicates of the input audio. We observed the model consistently train and evaluate well under our scale-adjusted training hyperparameters and an equivalent architecture to Hawthorne et al.

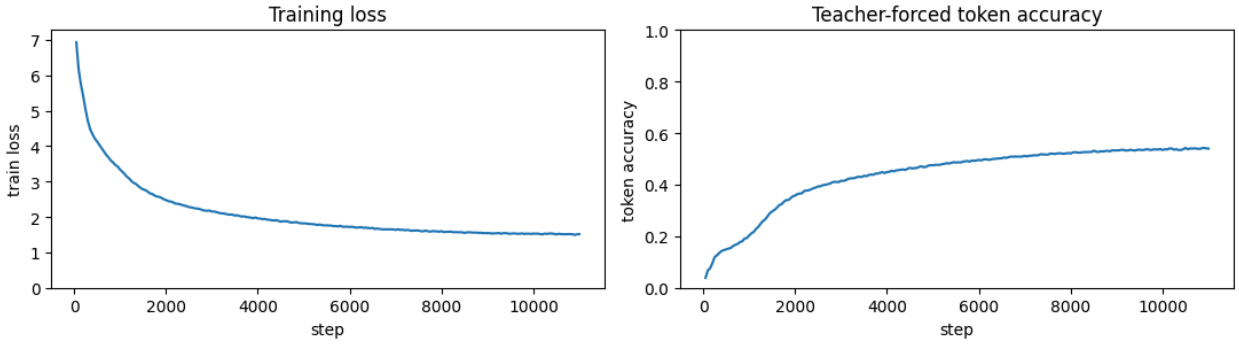

Fig 1. Training loss and teacher-forced token accuracy. During training, instead of letting the model use its own previous predictions as context (which could spiral if wrong), we force-fed it the *ground truth* tokens at each step. The accuracy here measures how often the model correctly predicts the next token under that setting.

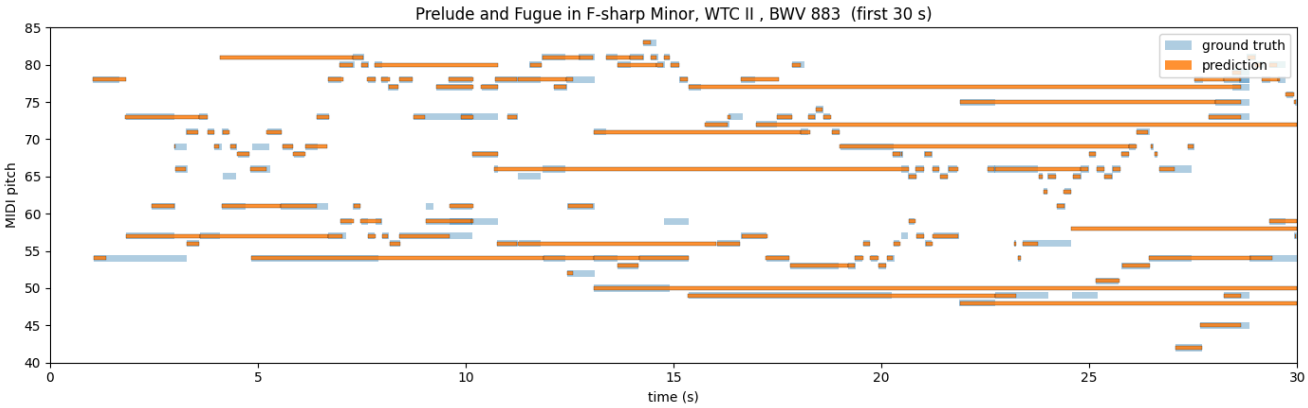

Fig 2. A comparison of original vs. transcribed audio (as MIDI) for a single piece in our test dataset

We also attempted to test our model on both the MAESTRO dataset, from which our training subset came, and the newer MAPS dataset, which crucially contains synthetic piano recordings—MAESTRO only contains real performances. To our surprise, the performance on both sets was very similar, with the MAPS pieces performing mildly better for onsets, and a little worse for our other F1 scores. Crucially, this shows that the model generalizes well beyond MAESTRO for its intended task (piano audio transcription).

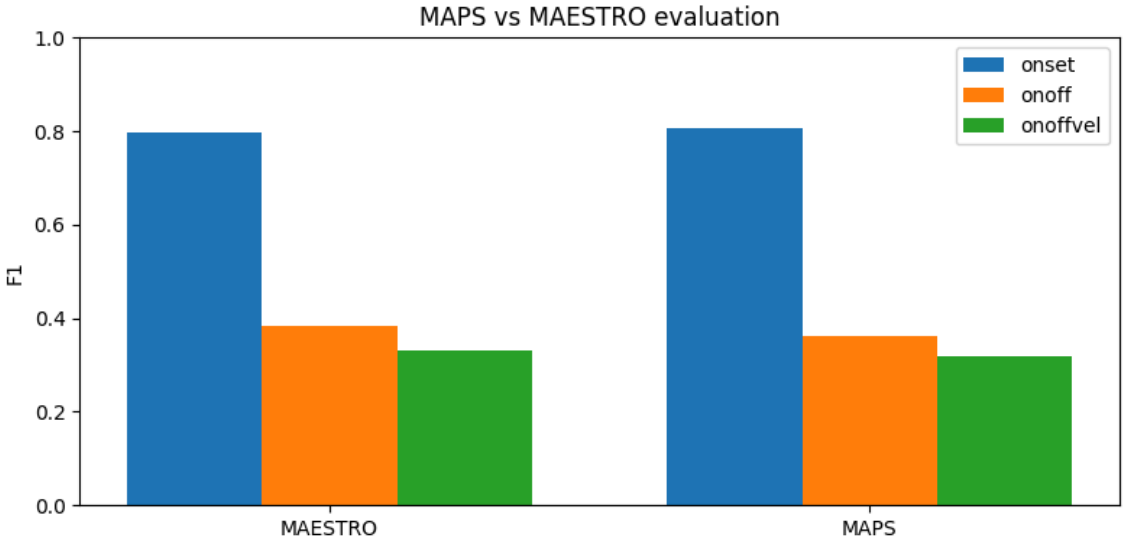

Fig 3. A comparison of test performance on MAESTRO (original training dataset) vs MAPS (new dataset).

During preprocessing, the MIDI files used as training examples are transformed into one-hot encoded MIDI-like events for the transformer. Part of the design behind this processing involves representing the gaps of time in the audio ("time shifts"). Hawthorne et al. claimed that representing time as an offset of units from the start of a segment, rather than a step-wise relative shift from the previous audio produces substantially better transcription performance. Our team hypothesized that this would hold in our experiment as the likely mechanism must persist in our replication: If time shifts are relative, each error hurts downstream correctness during decoding such that errors compound with sequence length.

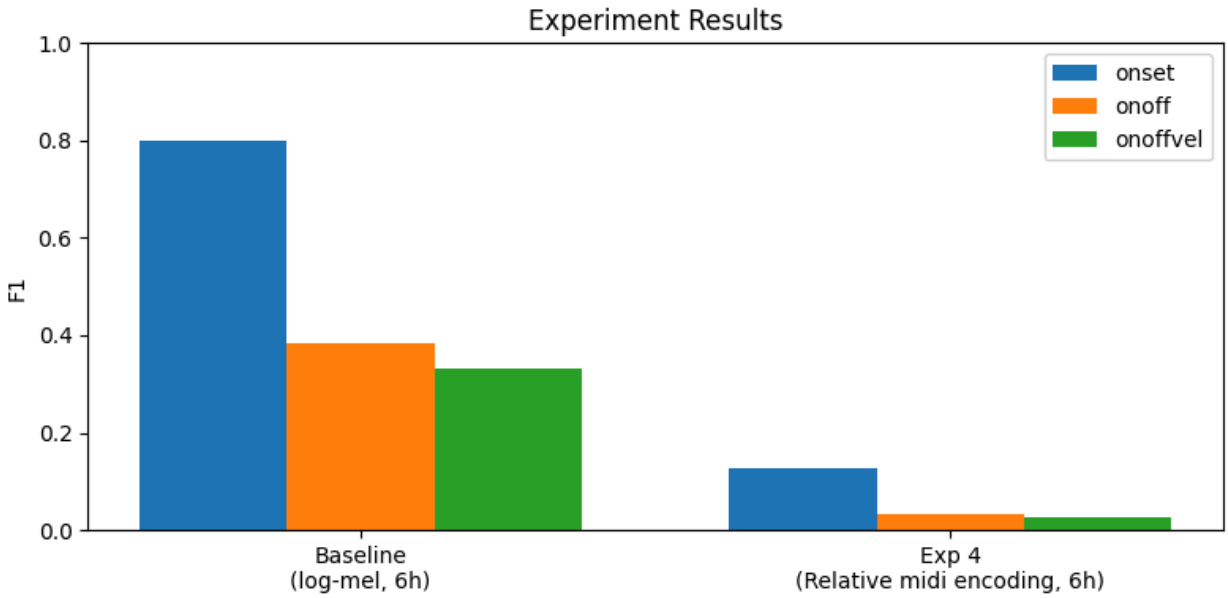

Fig 4. Both sides show each of our three described F1 score metrics (see methodology). The left side shows the performance of the original replicated model, whereas the right side shows the same model with absolute time shifts in the MIDI tokenizer swapped for relative time shifts.

As expected, this is exactly what occurred. In fact, relative time shifts performed substantially worse. We suspect that this is because our weakly trained model is already more likely to make per-event mistakes, which means over a similar sequence length as the paper, we see the compounding-error effect being more pronounced.

During preprocessing, raw spectrograms are transformed through two sequential steps: mel scaling, which maps frequencies to a scale that better matches how humans perceive pitch, and log scaling. Hawthorne et al. claimed that removing this combined log mel transformation allows successful training but causes poorer performance, attributing the importance to mel scaling producing useful features that the model would otherwise have to spend capacity extracting.

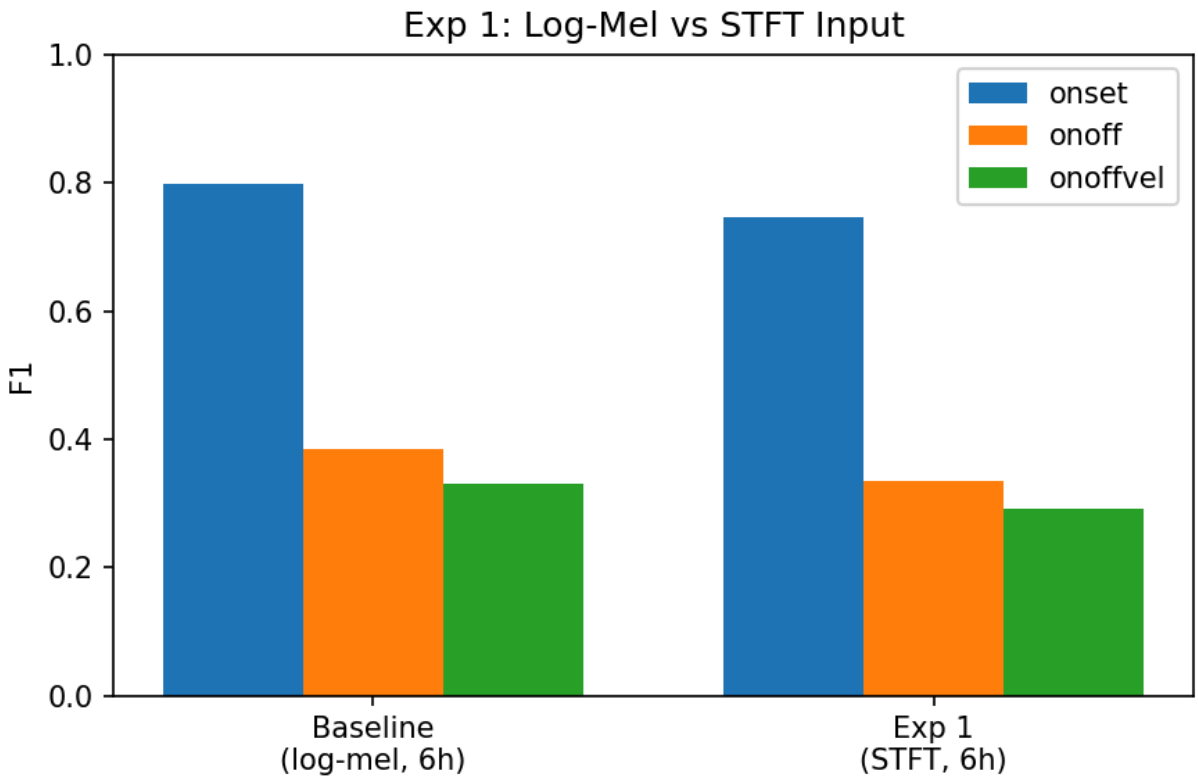

Fig 5. Both sides show each of our three described F1 score metrics (see methodology). The left side shows the performance of the original replicated model, whereas the right side shows the same model with log mel scaling omitted.

As expected, removing the log mel transformation did reduce performance. We suspect that in addition to the mechanism proposed by Hawthorne et al. it is possible that because the log scaling amplifies relative differences in small values (quiet sounds) while compressing relative

differences in large values (loud sounds), making percentage changes more uniform across the dynamic range, the model can better detect quiet notes that would otherwise be obscured by loud ones.

The architecture uses 6 attention heads per layer, following the original paper's design at full scale. However, we wanted to examine the effect of intentionally reducing the scale of the architecture by removing two heads—were all 6 heads really necessary for this AMT task? We hypothesized that this reduction would mildly reduce performance.

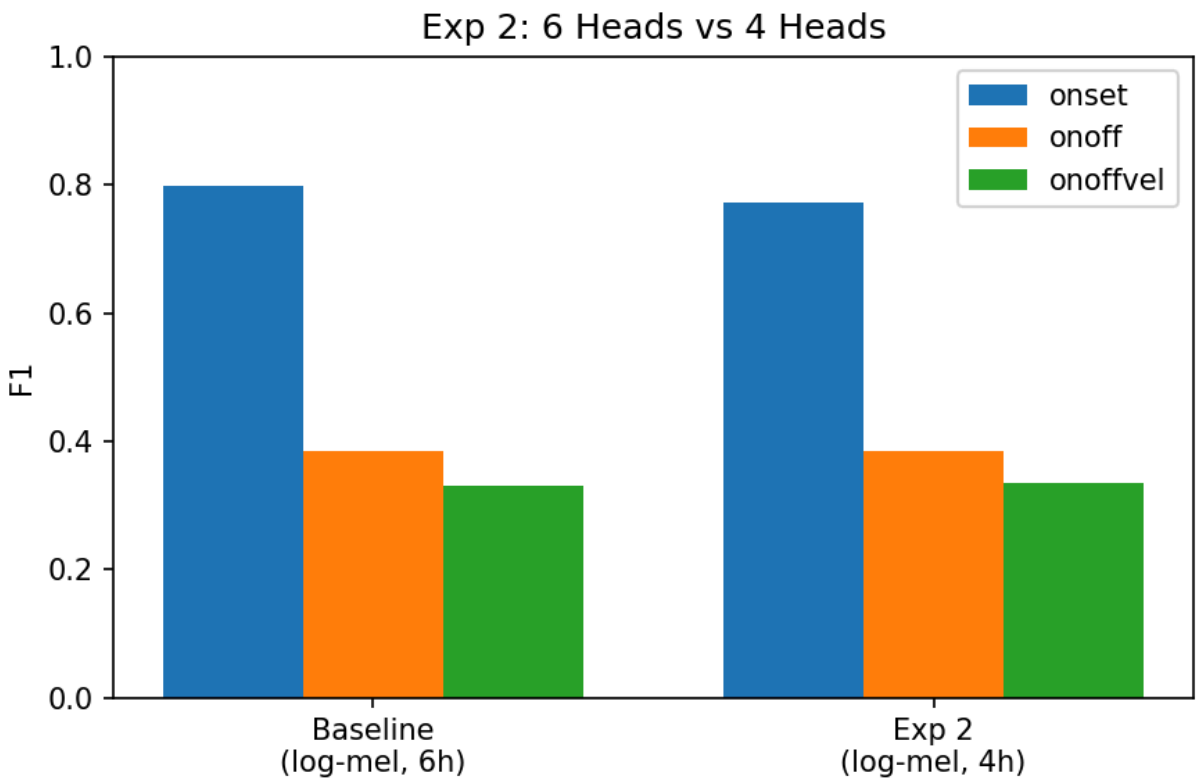

Fig 6. Both sides show each of our three described F1 score metrics (see methodology). The left side shows the performance of the original replicated model, whereas the right side shows the same model 2 transformer heads omitted.

As expected, removing 2 heads had very little impact on transcription quality across all F1

metrics. We argue that the lack of influence of the extra two heads is due to our lower-scale training regime: We could not provide enough data and training for meaningful six-way specialization.

The paper swaps T5's logarithmically scaled relative positional bucket embeddings for fixed absolute sinusoidal positional embeddings, arguing this ensures uniform attention resolution across all sequence positions. We hypothesized that reverting to T5's original relative bucket embeddings would meaningfully hurt performance.

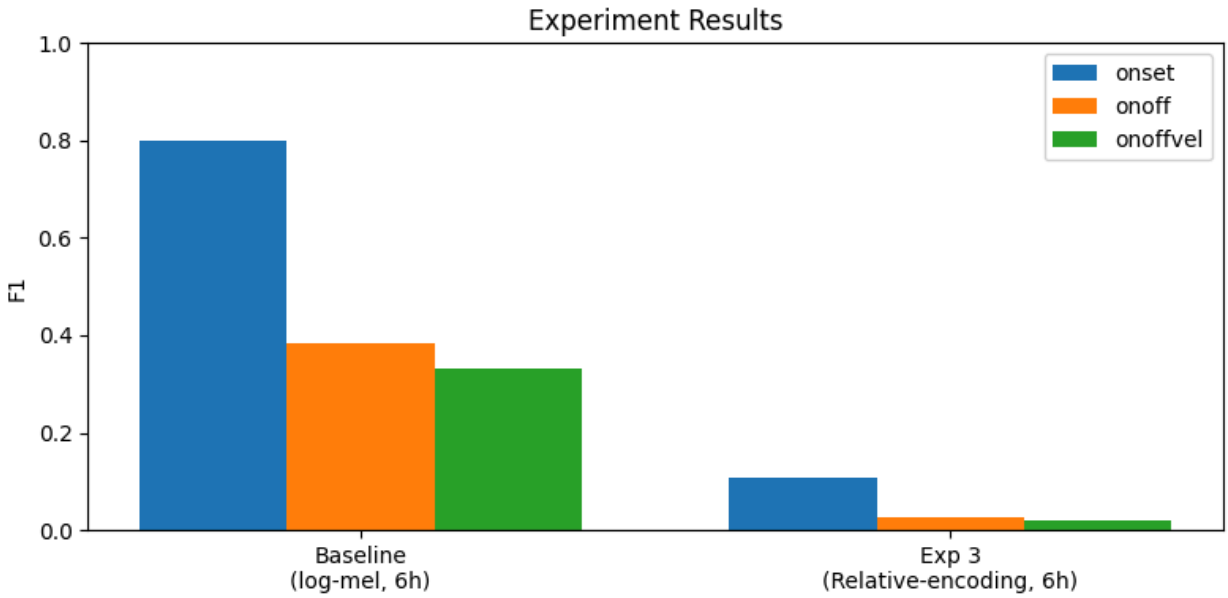

Fig 7. Both sides show each of our three described F1 score metrics (see methodology). The left side shows the performance of the original replicated model, whereas the right side shows the same model with T5-style relative positional bucket embeddings.

The results confirmed our hypothesis with significant performance degradation, particularly for onset+offset matching. We suspect this occurs because logarithmic bucketing assigns increasingly coarse-grained representations to distant context—a design appropriate for language

modeling where nearby tokens typically matter more, but problematic for music transcription. Unlike language, a note that started 300 frames ago has uniform importance to accuracy throughout its active duration, so the coarse representations for distant notes impair the model's ability to accurately track note offsets.

Challenges

There were a few challenges we encountered. The first part was just getting a complete, preprocessing, and model training pipeline to work. While the paper was simple to read, the codebase was a little complicated given it combined code from multiple papers. As we mentioned, the preprocessing with multiple datasets was also challenging. Once we had that ready, training the model was its own challenge given the compute limitations we have mentioned. Early on, we also dealt with issues of running out of memory during training, which we tackled by reducing the effective batch size. We also had a big issue with mode collapse at the early stages of training, where the model would predict a velocity of 0 very often, as this was the most common velocity. Eventually, this led us to increasing training steps. After 11k steps, we did see our loss reduce, and we produced a working model.

Ethics

One big flaw in our method is that our data is biased. Both MAESTRO and MAPS contain music that comes from a fairly homogenous time period and set of composers. Western classical music is very much overrepresented in the data. This means that the model has learned the patterns, time signatures, and rhythms of this type of music much more deeply than it has for music of other cultures. While we have not tested this, we suspect that running inference on music from other parts of the world or from other genres would therefore lead to poorer performance. This raises ethical concerns that would need to be addressed before the model is presented as a

product (instead of a research artifact). Beyond that, it is worth considering the environmental consequences that full scale training of a model like this entails. After all, Hawthorne et al. needed to run 32 GPUs over 2.5 days. Environmental ethics is one of the motivators of refining and evaluating this model and the work of Hawthorne et al. at a smaller scale.

Reflection

We are reasonably satisfied with how the project turned out. We did not reach stretch goals like going beyond replication, critical analysis, and stress testing of the architecture of Hawthorne at al. (e.g. extending the model with support for more instruments), we did clearly meet the target: All assignment components were completed on time, and our model demonstrated a practical level of success—the audio produced by the MIDI events generated by the model were perceptually convincing to human listeners.

Our original approach to the project was to center our focus on validating the underlying theme that generic architectures can match or exceed specialized, task-specific designs by comparing results between a similarly scaled domain-specific architecture and the transformer design we studied. Such an approach was poorly aligned with the assignment specifications, and the results would likely not have been particularly generalizable or insightful due to the sheer number of architecture design and hyperparameter selection decisions that would occur before a comparison could be made. If we could do this project again, we would consider an approach that is similarly ambitious to this, but better aligned and with more potential for creating insight in the deep learning space. If we had more time with our current approach, the quality of our work would benefit from longer training, generating a more diverse set of experimental metrics and visualizations, and trying other instruments with the same model architecture.

We had several significant takeaways and learning experiences by the end of the project. We

found that the model works surprisingly well even at 0.3% of the original compute, suggesting that testing architectures at smaller scales is not only effective, but also potentially valuable for reducing the need for computational resources. Secondly, we successfully expanded on the paper by rigorously evaluating the relative-to-absolute time encoding change, which the original paper mentioned but never validated—we found this choice significantly impacts performance and thereby gained a deeper understanding of the performance significance of tokenization during preprocessing. Thirdly, we gained domain expertise in audio processing; for example, we learned that because notes have sharp onsets but fade out gradually, it's harder for a model to learn offsets compared to onsets. Finally, we learned evaluation metrics beyond what was covered in class. In particular, we learned about F1 scores as a more sophisticated evaluation metric than "percent correct" and how teacher forcing during training can hide performance issues that emerge at test time.

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